

Problem statement No.40

Network Intrusion Detection

The Challenge

Network administrators face massive volumes of traffic and struggle to identify intrusions (like DoS, Probes, or Privilege Escalation) in real-time. Manually analysing flow logs and writing secure firewall rules is time-consuming and prone to human error, leaving networks vulnerable.

The Objective

Build an AI-powered Network Intrusion Detection System (NIDS) SOC Agent that delivers autonomous threat classification and mitigation for security teams. The solution should:

Real-time Traffic Analysis: Classify network flow metrics (e.g., NSL-KDD features) into threat categories.

Threat Intelligence Retrieval: Map identified attacks to the MITRE ATT&CK framework.

Automated Mitigation: Generate precise iptables firewall scripts to block malicious IPs.

Safety Guardrails: Validate generated rules against enterprise safety policies before deployment.

Multi-Agent System:

NIDS Classifier Agent (Retrieves ML predictions)

Threat Intel Agent (Fetches MITRE context)

Remediation Agent (Writes firewall scripts)

Policy Guard Agent (Validates scripts for safety)

